

Fiscal Policy and Sectoral Spillovers

Two-sector open-economy HANK

Seminar notes on de Beaufort and Rannenberg (2026)

Open-source notes on NBB Working Paper No. 493
not the authors' replication package

`de-beaufort-rannenberg/` in MacroModellerBot

September 2026

Source paper: de Beaufort and Rannenberg (2026), NBB WP 493 (June 2026),
“Fiscal policy and sectoral spillovers in open-economy HANK.”

Outline

- 1 Literature
- 2 The model
- 3 Contributions
- 4 Results and discussion
- 5 What the repo prototype achieves
- 6 Simplifying assumptions
- 7 Extensions

Two tracks, in that order: the *paper* (literature, two-sector open HANK, claims), then what this repo's Python / sequence-jacobian *prototype* actually does.

Why fiscal non-tradables?

Core question de Beaufort and Rannenberg (2026)

Government spending falls disproportionately on *non-tradable services*. Empirically, a spending shock still raises private consumption (goods more than services), raises goods *and* services output, *lowers* the relative price of goods, and *worsens* net exports.

- Nested two-sector open RANK with the same G bias does *not* deliver that package: C falls, and when G is purely NT the goods-output co-movement disappears.
- The paper asks whether **household heterogeneity** plus **trade openness** can generate the consumption-led spillover from fiscal non-tradables into the goods sector.
- Object: **sectoral fiscal multipliers** in an open economy—not a one-sector closed-economy multiplier, and not transfers/FTPL.

Sectoral fiscal puzzles: RANK, TANK, open HANK

- **Open RANK, sectoral.** Monacelli and Perotti (2008, 2010): C falls in both sectors; when G is services-intensive, Y_T contracts (negative output co-movement). Openness can help relative prices, not that package. This paper adds NX and a pre-COVID sample.
- **Fiscal HANK / IKC.** High iMPCs reverse RANK crowding-out of C (Galí, López-Salido and Vallés 2007 TANK; Auclert, Rognlie and Straub 2024 IKC—the household block here). Sticky information (Auclert, Rognlie and Straub 2020; $\vartheta = 0.935$) for humps; signs survive under RE.
- **Open fiscal HANK, one sector.** Druedahl et al. (2024): import leakage offsets part of the HANK multiplier. Aggarwal, Auclert, Rognlie and Straub (2023): high iMPCs $\Rightarrow$ persistent CA deficits. Missing piece: T/NT relative prices.
- **T / NT structure.** Cardi, Restout and Claeys (2020): imperfect labour mobility after NT-biased G . Corsetti and Dedola (2005): distribution costs in retail tradable prices. Composition of G then matters.

Contrast, not synthesis: Kaplan–Miyahara (NBER WP 35400) and Angeletos, Lian, Wolf and Zhang (NBER WP 35642) are HANK *monetary–fiscal / transfer* papers. This talk is NT-biased G and sectoral multipliers.

- **RANK.** Ricardian Euler. Deficit-financed G raises expected real rates *and* the PV of taxes, so C falls in both sectors (Monacelli–Perotti). $Y_T > 0$ then rides on the tradable content of G ; with $\phi_{GT} = 0$, that co-movement vanishes. NX barely moves.
- **HANK.** Uninsurable income risk and $a \geq 0$ produce high iMPCs and precautionary saving. NT-biased G raises labour income; constrained households spend it, and CES demand spills into goods—even if $\phi_{GT} = 0$.
- **Openness is not a sideshow.** Import intensity plus imperfect labour reallocation dampen wage pass-through to tradable prices. P_T/P_{NT} can *fall* as Y_T rises; NX can deteriorate while expenditure switching still supports Y_T .
- **Paper's identification of the mechanism.** Nested RANK vs. HANK, and a $\phi_{GT} = 0$ (pure NT) counterfactual: HANK keeps $Y_T > 0$; RANK does not.

Four stylised facts (S-BVAR)

Empirical block (WP §2): Blanchard–Perotti, G ordered first (Cholesky), U.S. 1954Q1–2019Q4, nine variables, four lags, NIW prior. Shock: orthogonal **1% of output** increase in G .

- ➊ Private consumption **rises** (crowding-in), with C_T **larger** than C_{NT} .
- ➋ Output rises in **goods and services** (despite NT-biased G).
- ➌ The relative price of goods **falls**.
- ➍ Net exports **deteriorate**.

Ramey-style military narrative is a poor proxy here (more tradable; anticipation). The theoretical G path is the empirical IRF (“does not follow a hump-shaped path”, §4.3). This repo uses a 20-quarter path digitised from Figure 3’s G panel [**PROVISIONAL**], not the authors’ series.

Two-sector open HANK at a glance

- Domestic production: **tradables** (T , goods) and **non-tradables** (NT , services). Linear technology $Y_S = N_S$; flexible goods prices, sticky wages only (baseline).
- Incomplete-markets households (Auclert–Rognlie–Straub 2024); hours set by sectoral unions ($n_{S,t}(i) = N_{S,t}$). Nested RANK: same aggregates, Euler instead of the HA Jacobian.
- Fiscal G is CES-split with $\phi_{GT} = 0.25$ (OECD COFOG tradable content); HA-LB / RA-LB set $\phi_{GT} = 0$.
- Open-economy *block* with foreign variables at SS, calibrated to the U.S. (12% import/GDP): CES home/foreign, Corsetti–Dedola distribution, UIP with an NFA wedge, tax rule on debt, Taylor on CPI.
- Solution: perfect-foresight MIT shocks, sequence-space Jacobians, linearised (Auclert, Bardóczy, Rognlie and Straub 2021).

- Incomplete markets, $a \geq 0$, idiosyncratic e_t AR(1) ($\rho_e = 0.968$). Post-tax labour income

$$z_t(i) = (w_T N_T + w_{NT} N_{NT} - T_t) e_t(i).$$

- Utility (4) with $\sigma = \varphi = 1$. Households choose c, a only; hours are union-set. Taxes scale with e , so the G shock does not reshape the income *distribution*.
- CES demands: T vs. NT ($\lambda_t = 2.15$, $\phi_t = 0.623$, including a distribution-margin reclassification); home vs. foreign with trade elasticity $\lambda_d = 1$ (Cobb–Douglas; 12% import/GDP via $\phi_d = 0.57$).
- Sticky expectations, share $\vartheta = 0.935$ that does not update (paper eq. 13 / Auclert–Rognlie–Straub mapping). Used for *humps*; qualitative signs are not tied to it.
- Nested RANK: same budget, representative Euler, same r ($\beta_{RA} = 0.995$ vs. $\beta_{HA} = 0.986$).

- Flexible prices: $P_Z = W_S$ (eq. 14). Sticky-price Appendix C is not in the baseline DAG (avoids countercyclical profits).
- Imperfect labour mobility: Horvath aggregator (9), $\lambda_\ell = 1$. Sectoral wage Phillips curves (12), slope $\kappa_w = 0.0044$ (Lindé et al. 2016), $\theta_w = 3$ (50% markup).
- Two reasons P_T/P_{NT} falls: (i) the foreign component of tradable prices; (ii) NT-biased G plus limited reallocation, so NT wages rise more.
- Distribution (Corsetti–Dedola, eqs. 28–31): local NT services enter retail prices of domestic and imported tradables ($\mu_d = 0.266$, $\mu_f = 0.754$). Import NT content $\approx 2\times$ domestic, which *cushions* NX after appreciation.
- Output identities (34)–(35): Y_T is domestic tradables plus exports; Y_{NT} is services plus distribution margins.

$$T_t = T_{ss} + \kappa_b(B_{t-1} - B_{ss}) \quad (\text{tax, 21})$$

$$i_t = \zeta i_{t-1} + (1 - \zeta)(i^* + \kappa_\pi \pi_t) \quad (\text{Taylor, 22})$$

- G exogenous; CES split (16)–(19) with $\phi_{GT} = 0.25$, $\phi_d^G = 1$ (no public import leakage in baseline). Budget (20). $\kappa_b = 0.027$, $\zeta = 0.9$, $\kappa_\pi = 1.5$.
- Mutual fund / UIP (23)–(26). Paper: $\Gamma_t = \exp(-10^{-9} \text{nfa}_t)$ (Schmitt-Grohé–Uribe NFA closer).
- Exports $X_t = C_{F,t}^*/(1 + \mu_f)$; imports through the distribution wedge. Goods/asset clearing (36)–(37).
- SSJ DAG: unknowns $(Y_T, Y_{NT}, w_T, w_{NT}, \pi, Q, i, B, \text{nfa})$; goods_NT dropped by Walras (household + government + NFA laws of motion).

Table 2 (paper) vs. what we feed the computer

Param. (Table 2)	Value	Param.	Value
β_{RA}, β_{HA}	0.995, 0.986	ϕ_t, λ_t	0.623, 2.15
θ_w, κ_w	3, 0.0044	ϕ_d, λ_d	0.57, 1
σ, φ	1, 1	μ_f, μ_d	0.754, 0.266
ρ_e, ϑ	0.968, 0.935	ζ, κ_π	0.9, 1.5
ϕ_{GT}, κ_b	0.25, 0.027	λ_ℓ, ϕ_d^G	1, 1

- All of the above are **[VERIFIED]** in TABLE2.
- **[PROVISIONAL]** (NOTES.md; later slide): $\sigma_e = 0.35$ rather than Table 2's 0.58; $\psi_{\text{nfa}} = 10^{-3}$ rather than 10^{-9} ; G/Y and ϕ_ℓ implied by adding-up.
- Estimation of the S-BVAR itself is **not** in this repo.

What the NBB WP adds

Relative to one-sector HANK and to open-economy RANK fiscal:

- An **empirical** four-fact package, including NX and a $C_T > C_{NT}$ split, for a G shock concentrated in NT services (S-BVAR through 2019Q4).
- A **two-sector open HANK**: uninsurable income risk and precautionary saving produce a persistent income-driven expansion in C ; import intensity and limited labour reallocation dampen wage pass-through to goods prices.
- **Expenditure switching** then yields $Y_T > 0$ even as NX worsens—and even when $\phi_{GT} = 0$. Nested RANK fails that test. TANK (App. B) can raise C on impact but lacks the precautionary hump.
- Quantitative message: **household heterogeneity and trade openness jointly** shape sectoral fiscal multipliers. Composition of G is not a sideshow; more open economies leak more of Y_T (App. D).

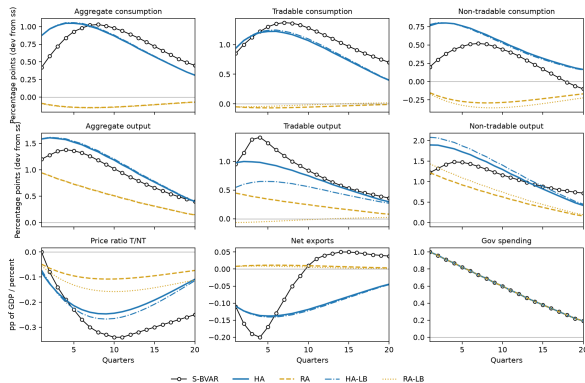
What this open-source prototype adds

Honest split vs. the published paper:

- **Runnable Python + sequence-jacobian** two-sector open HANK / nested RANK DAG, linear MIT-shock IRFs, and a Figure 3 layout (HA, RA, HA-LB, RA-LB, digitised S-BVAR overlay).
- Table 2 parameters in `calibration.py`; qualitative sign checks in `test_qualitative.py` / `figure3_summary.txt`.
- Documented **[PROVISIONAL]** closures needed to complete the DAG (next slides): σ_e , UIP wedge, digitised G / S-BVAR, foreign export demand, distribution on G_D .
- **Does not** reproduce author IRF numbers, S-BVAR 68% bands, Figure 4 international panel (as the main artefact), TANK App. B, or Appendices C–D. No official replication package / S-BVAR data were found.

Prototype IRF: Figure 3 layout

Impulse responses to a government spending shock (WP 493 Figure 3 layout)



S-BVAR markers are hand-digitised from WP 493 Figure 3 [PROVISIONAL], not the authors' numerical series.

Python + sequence-jacobian 1.0.0 prototype; 1% of GDP G shock, empirical-shaped path; $\sigma_e = 0.35$, $\psi_{\text{infa}} = 10^{-3}$.
S-BVAR markers are hand-digitised from WP 493 Figure 3. **[PROVISIONAL]** not the authors' numerical IRFs.

Qualitative co-movements (this code)

Signs from output/figure3_summary.txt (near-term averages), matching the paper's main *message*—not author magnitudes:

Fact	HA	HA-LB	RA	RA-LB
$C > 0$	yes	yes	no (falls)	no
$Y_T > 0$	yes	yes	yes (from ϕ_{GT})	no
$Y_{NT} > 0$	yes	yes	yes	yes
P_T/P_{NT} down	yes	yes	yes (smaller)	yes
NX down	yes	yes	no (tiny +)	no
$C_T > C_{NT}$ response	yes	yes	yes	yes

- Heterogeneity delivers the consumption-led spillover to tradables; RANK does not, especially when G is purely non-tradable.
- RA still has $Y_T > 0$ at $\phi_{GT} = 0.25$: that is G 's tradable content, not crowding-in of C .

Mechanism, and what is *not* claimed

- **HANK mechanism (paper).** NT-biased $G \rightarrow$ labour income $\rightarrow$ high iMPC consumption $\rightarrow$ CES demand for goods $\rightarrow Y_T > 0$ even if $\phi_{GT} = 0$. Distribution + imperfect mobility: P_T/P_{NT} can fall. Openness: NX deteriorates while expenditure switching supports Y_T .
- **RANK contrast.** C falls (higher r and future taxes). With $\phi_{GT} = 0.25$, Y_T can still rise from the tradable content of G ; with $\phi_{GT} = 0$, that co-movement disappears.
- **Known shape gaps vs. the published figure** (NOTES.md; this code): HA C is a bit too front-loaded (sticky-info hump milder); HA Y_T does not reproduce the empirical hump; HA NX stays negative too long (the authors note a medium-run NX undershoot); RANK NX is a small surplus rather than a small deficit.

Why this matters for recent fiscal / open-economy debates

- **Composition of G .** Health, education, administration are services. Multipliers quoted on aggregate G hide a T/NT split; $\phi_{GT} = 0.25$ is the empirically relevant bias, not a knife-edge.
- **Crowding-in vs. crowding-out.** The S-BVAR sample ends in 2019Q4. The *mechanism*—high iMPCs plus NT-biased G —is what travels to later packages, not a COVID-period estimate. Kaplan–Miyahara / Angeletos et al. speak to transfers and monetary–fiscal inflation; different object.
- **Openness and leakage.** At U.S. import intensity, expenditure switching can raise Y_T even as NX worsens. Appendix D: a *more* open economy leaks more of that goods-sector gain. Closed-economy HANK misses the joint restriction.
- **Relative prices.** NT-heavy G can *lower* goods prices relative to services—a sectoral relative-price fact, not a one-sector inflation multiplier.

Runs today (Python + sequence-jacobian 1.0.0; Table 2 params unless noted):

- Two-sector open HANK (sticky-info HA) and nested RANK.
- $\phi_{GT} = 0.25$ baseline and $\phi_{GT} = 0$ (LB) variants.
- Figure 3 PNG + 20-quarter CSVs + qualitative summary.
- Smoke test of HA signs, RANK $C < 0$, and RA-LB $Y_T \not\approx 0$.

```
cd de-beaufort-rannenberg
python3 src/run_figure3.py
python3 src/run_figure3.py --sigma-e 0.35 --psi-nfa 1e-3
python3 src/run_figure3.py --T 80 --skip-lb # faster smoke
python3 src/test_qualitative.py
```

Writes output/figure3.png, irf_{ha,ra,ha_lb,ra_lb}.csv, figure3_summary.txt.

Does not run: author S-BVAR estimation, exact G vector, Appendix B TANK, Appendix C sticky prices, Appendix D 25% import share.

Calibration knobs and caveats

- σ_e . Table 2 lists 0.58 (Auclert et al. 2024 quarterly conversion). As the SSJ Rouwenhorst SD of $\log e$, that undershoots Table 1 year-1 iMPC 0.507 and text HtM 49.7% ($\approx 0.24 / \approx 20\%$). Default `--sigma-e 0.35` gets $iMPC_1 \approx 0.53$, HtM $\approx 49.5\%$ given β_{HA} and $r = 1/\beta_{RA} - 1$ [PROVISIONAL]—close, not exact.
- **UIP wedge**. Paper $\psi_{nfa} = 10^{-9}$; CLI default `--psi-nfa 1e-3` for a stable linear solve [PROVISIONAL].
- **S-BVAR overlay** in `empirical_sbvar.py`: hand-read from Figure 3, not the authors' series [PROVISIONAL].
- **Other closures** (NOTES.md): $G/Y \approx 0.214$ implied by 12% import/GDP; ϕ_ℓ set to sectoral hours shares so both wage PCs hold at $w = 1$; distribution margin on G_D for labour income = GDP; SOE export demand with C^* fixed; SSJ ex-post r = lagged ex-ante.

What is on, what is shut off

In the paper's baseline:

- Two domestic sectors only; linear technology; flexible prices / sticky wages (not sticky prices).
- Hours set by unions; incomplete markets with $a \geq 0$; sticky expectations on the HA Jacobian (signs not tied to it).
- Foreign variables at SS; Cobb–Douglas trade ($\lambda_d = 1$); Horvath $\lambda_\ell = 1$; $\phi_{GT} = 0.25$; $\phi_d^G = 1$.

Extra prototype approximations (not author codes):

- $\sigma_e = 0.35$ vs. Table 2 0.58; $\psi_{\text{nfa}} = 10^{-3}$ vs. 10^{-9} ; digitised G and S-BVAR; provisional foreign CES / distribution on G_D ; SSJ r timing.

Shut off in both (baseline script): Appendix B TANK; Appendix C sticky prices; Appendix D 25% import share; Figure 4 international panel as the main artefact; official S-BVAR data.

Future developments that fit this codebase

- 1 **N-sector sketch** (EXTENDING.md / sectors.py). Today DOMESTIC_SECTORS = (T, NT); blocks are still explicit two-sector objects. Generate per-sector wage PC / hours / demand from that tuple instead of copy-pasting T/NT .
- 2 **Nesting, not a monolith**. A third sector S needs an explicit CES / distribution / tradable status. Keep trade, UIP, and the HA Jacobian as separate SSJ blocks. Walras: still drop *exactly one* sectoral goods residual.
- 3 **Paper appendices** (DAG is extensible). TANK (App. B); sticky prices (App. C); 25% import share (App. D); Figure 4 international panel as a first-class artefact.
- 4 **Blocked on data/codes**. Official replication files, S-BVAR data, exact G IRF vector, fully specified foreign block.
- 5 **Later, not this folder**. HA-OLG household block: Bardóczy–Shanker–Velásquez–Giraldo (FEDS 2026-046) is the SSJ cookbook—unused here (infinite horizon). Synthesis with the HENK companion: two-sector open HANK fiscal plus heterogeneous-expectation NK.

- NT-biased government spending can still raise C , raise Y_T and Y_{NT} , lower P_T/P_{NT} , and worsen NX —if households are incomplete-markets and the economy is open.
- Nested RANK fails that package, especially when $\phi_{GT} = 0$. Heterogeneity, not the tradable content of G , is the spillover.
- This repo is a transparent Figure 3 starter: Python + SSJ DAG, qualitative HANK vs. RANK signs, documented provisional σ_e / UIP / S-BVAR closures—not a numerical replica of the authors' IRFs.
- The natural extension is a labelled multi-sector open DAG, not a one-file N-sector solver.

Selected references. de Beaufort and Rannenberg (2026), NBB WP 493; Monacelli and Perotti (2008, 2010); Auclert, Rognlie and Straub (2020, 2024); Auclert, Bardóczy, Rognlie and Straub (2021); Druedahl et al. (2024); Aggarwal et al. (2023); Cardì, Restout and Claeys (2020); Corsetti and Dedola (2005); Galí, López-Salido and Vallés (2007). Compile: `latexmk -pdf dbr_seminar.tex` from `de-beaufort-rannenberg/beamer/`.